TUMORS OF ENT

ENT | 3rd Year MBBS

Hi Sammy!
ENT tumors = high yield in university!
Learn the classics!

Tumors | NPC | JNA | Laryngeal CA | ENT

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JNA = adolescent male
with epistaxis!
Never biopsy it!

Nasopharyngeal Carcinoma (NPC)

Histology:	Squamous cell carcinoma (undifferentiated = lymphoepithelioma)
Association:	Epstein-Barr virus (EBV) — elevated anti-VCA IgA titers
Geography:	Common in Southern China, SE Asia (endemic regions)
Site:	Fossa of Rosenmuller (pharyngeal recess) — most common site
Presentation:	Neck mass (cervical LN metastasis), nasal obstruction, epistaxis
Trotter's triad:	Unilateral conductive deafness + palatal asymmetry + trigeminal neuralgia

Juvenile Nasopharyngeal Angiofibroma (JNA)

Nature:	Benign but locally aggressive vascular tumor
Who:	Exclusively in ADOLESCENT MALES (10-25 years)
Origin:	Sphenopalatine foramen (superior margin)
Symptoms:	Recurrent unilateral epistaxis + progressive nasal obstruction
Signs:	Frog face deformity (if extending into cheek), proptosis
Diagnosis:	CT/MRI with contrast (Holman-Miller sign on XR), CT angiography
NEVER biopsy:	Highly vascular — biopsy causes torrential bleeding!

JNA Treatment

- Pre-op embolization (to reduce vascularity) -> surgical excision
- Lateral rhinotomy / midfacial degloving / endoscopic approach
- Radiotherapy for recurrence or intracranial extension only

JNA Classic Viva Q!
Adolescent male + recurrent epistaxis + nasal mass = JNA until proven otherwise. NEVER biopsy!

Glottic vs supraglottic
= THE comparison in
laryngeal carcinoma!

Laryngeal Carcinoma — Overview

- Most common malignancy of larynx = Squamous Cell Carcinoma (SCC > 95%)
- Risk factors: Smoking (#1), alcohol, GERD, HPV, asbestos, radiation
- Males >> Females (10:1 ratio). Peak age: 50-70 years

Glottic vs Supraglottic vs Subglottic

Feature	Glottic	Supraglottic	Subglottic
Frequency	60% (most common)	35%	5% (rarest)
Early symptom	Hoarseness (EARLY)	Dysphagia, foreign body	Stridor (late)
LN mets	RARE (no lymphatics)	COMMON (rich lymph.)	Moderate
Prognosis	BEST (early detection)	Intermediate	WORST (late detection)
Staging	Vocal cord mobility	Pre-epiglottic space	Subglottic extension
Treatment	RT / cordectomy	RT +/- surgery	Laryngectomy + RT

★ **Glottic CA has the BEST prognosis because hoarseness appears EARLY**
and the glottis has minimal lymphatic drainage = rare metastasis!

Clinical Features of Laryngeal Carcinoma

- **Hoarseness** — Earliest & most common symptom (especially glottic)
- **Dysphagia** — Supraglottic tumors — referred otalgia via CN IX
- **Stridor** — Late sign — indicates significant airway obstruction
- **Neck mass** — Cervical lymphadenopathy (supraglottic > subglottic)
- **Weight loss** — Advanced disease, cachexia
- **Hemoptysis** — Blood-streaked sputum — alarming sign

Treatment Principles

1. Early glottic: Radiotherapy or laser cordectomy (voice preservation)
2. Advanced: Total laryngectomy + neck dissection + post-op RT
3. Voice rehabilitation: Tracheoesophageal puncture (TEP), esophageal speech

Acoustic neuroma =
most common CPA tumor!
MRI is the gold std!

Acoustic Neuroma (Vestibular Schwannoma)

Nature:	Benign tumor of Schwann cells of CN VIII (vestibular division)
Location:	Cerebellopontine angle (CPA) — most common CPA tumor
Bilateral:	Bilateral acoustic neuromas = Neurofibromatosis type 2 (NF2)
Presentation:	Unilateral progressive SNHL + tinnitus + imbalance
CN involvement:	CN V (numbness), CN VII (facial palsy) in large tumors
Diagnosis:	MRI with gadolinium contrast (gold standard)
Treatment:	Observation (small) / stereotactic radiosurgery / microsurgery

Sinonasal Tumors

Tumor	Key Features
Inverted papilloma	Benign but locally aggressive, high recurrence, may turn malignant
SCC maxillary sinus	Most common malignant sinonasal tumor; Ohngren's line for staging
Esthesioneuroblastoma	Arises from olfactory epithelium; Kadish staging; bimodal age
Adenoid cystic CA	Perineural invasion! Slow-growing, late recurrence, minor salivary gland
Melanoma	Sinonasal melanoma — very aggressive, poor prognosis
NK/T cell lymphoma	Nasal type; associated with EBV; lethal midline granuloma

★ **Ohngren's Line!**
Imaginary line from medial canthus to angle of mandible. Divides maxillary sinus:
Superoposterior (infrastructure) = WORSE prognosis; Anteroinferior = BETTER prognosis.

Other Important ENT Tumors

Tumor	Key Features
Glomus jugulare	Paraganglioma, pulsatile tinnitus, rising sun sign on otoscopy
Pleomorphic adenoma	Most common benign parotid tumor (80%); mixed tumor
Warthin tumor	2nd most common benign parotid; bilateral, smokers, males
Mucoepidermoid CA	Most common malignant salivary gland tumor
Laryngeal papilloma	HPV 6,11; recurrent respiratory papillomatosis in children

TNM staging comes in every surgery + ENT exam paper!

TNM Staging — Quick Reference

1. T = Tumor size and local invasion (T1-T4)
2. N = Regional lymph node involvement (N0-N3)
3. M = Distant metastasis (M0 or M1)

Diagnostic Workup for ENT Tumors

Investigation	Purpose
Direct laryngoscopy	Visualization + biopsy of laryngeal tumors
Nasal endoscopy	Assessment of nasopharyngeal / sinonasal tumors
CECT neck	Tumor extent, lymph node assessment, invasion
MRI	Soft tissue detail, perineural spread, intracranial extension
PET-CT	Distant metastasis, recurrence detection
FNAC / biopsy	Histological confirmation (NOT for JNA!)
EBV serology	Anti-VCA IgA for nasopharyngeal carcinoma

Treatment Summary — Key ENT Tumors

Tumor	Primary Treatment	Notes
NPC	Chemoradiation (RT+CT)	Very radiosensitive; surgery limited role
Laryngeal (early)	RT / laser surgery	Voice preservation goal
Laryngeal (advanced)	Total laryngectomy + RT	TEP for voice rehabilitation
JNA	Surgery (post-embolization)	Never biopsy! Pre-op embolization
Acoustic neuroma	Surgery / radiosurgery	MRI follow-up if observing
Maxillary SCC	Maxillectomy + RT	Infrastructure = better prognosis

Neck Dissection — Types

1. Radical: Removes LN levels I-V + SCM + IJV + CN XI (spinal accessory)
2. Modified radical: Preserves 1+ structures (SCM/IJV/CN XI)
3. Selective: Only specific levels removed based on tumor drainage

ENT tumors = 15+ marks potential!
You've GOT this Sammy!

Most Repeated LAQs (10–12 marks)

1. Describe JNA — features, diagnosis & management. Why not biopsy?
2. Compare glottic vs supraglottic carcinoma with management
3. Describe NPC — etiology, presentation, Trotter's triad & treatment
4. Acoustic neuroma — clinical features, diagnosis & management
5. Describe maxillary sinus carcinoma with Ohngren's line significance

Most Repeated SAQs (5–6 marks)

1. Trotter's triad & its significance
2. JNA — features, Holman-Miller sign, why no biopsy
3. Ohngren's line & prognostic significance
4. Types of neck dissection
5. Glomus jugulare — features & diagnosis
6. Inverted papilloma — features & management

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Adolescent male + epistaxis + nasal mass	JNA (never biopsy!)
EBV-associated nasopharyngeal tumor	Nasopharyngeal carcinoma
Fossa of Rosenmuller	Most common site of NPC
Trotter's triad	Deafness + palate asymm. + CN V neuralgia
Most common site of laryngeal CA	Glottis (60%)
Hoarseness > 3 weeks in smoker	Rule out laryngeal carcinoma
Most common CPA tumor	Acoustic neuroma (vestibular schwannoma)
Bilateral acoustic neuromas	Neurofibromatosis type 2 (NF2)
Rising sun sign on otoscopy	Glomus jugulare
Most common benign parotid tumor	Pleomorphic adenoma
Perineural invasion tumor	Adenoid cystic carcinoma
Holman-Miller sign (antral bowing)	JNA on X-ray

♥ **Early detection saves lives, Sammy! You'll be that alert doctor!**
Every tumor you diagnose early is a life you change forever.